Сборка производилась на Linux Mint 22.3, процессор Intel Core i5-12400F (6 ядер по 2 потока на каждый), архитектура amd64.

Перед началом сборки установим кросс-компилятор под arm:

```
vadim@vadim-PC:~$ sudo apt install gcc-arm-linux-gnueabihf
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils-arm-linux-gnueabihf cpp-13-arm-linux-gnueabihf cpp-arm-linux-gnueabihf
  gcc-13-arm-linux-gnueabihf gcc-13-arm-linux-gnueabihf-base gcc-13-cross-base gcc-14-cross-base
  libasan8-armhf-cross libatomic1-armhf-cross libc6-armhf-cross libc6-dev-armhf-cross
  libgcc-13-dev-armhf-cross libgcc-s1-armhf-cross libgomp1-armhf-cross libstdc++6-armhf-cross
  libubsan1-armhf-cross linux-libc-dev-armhf-cross
Suggested packages:
  binutils-doc gcc-13-locales cpp-13-doc cpp-doc gcc-13-doc gdb-arm-linux-gnueabihf gcc-doc
The following packages will be REMOVED:
  g++-multilib gcc-multilib
The following NEW packages will be installed:
  binutils-arm-linux-gnueabihf cpp-13-arm-linux-gnueabihf cpp-arm-linux-gnueabihf
  gcc-13-arm-linux-gnueabihf gcc-13-arm-linux-gnueabihf-base gcc-13-cross-base gcc-14-cross-base
  gcc-arm-linux-gnueabihf libasan8-armhf-cross libatomic1-armhf-cross libc6-armhf-cross
  libc6-dev-armhf-cross libgcc-13-dev-armhf-cross libgcc-s1-armhf-cross libgomp1-armhf-cross
  libstdc++6-armhf-cross libubsan1-armhf-cross linux-libc-dev-armhf-cross
0 upgraded, 18 newly installed, 2 to remove and 0 not upgraded.
Need to get 43,3 MB of archives.
After this operation, 138 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://mirror.yandex.ru/ubuntu noble-updates/main amd64 gcc-13-arm-linux-gnueabihf-base amd6
```

Создаём конфигурацию по-умолчанию под ARM:

```
vadim@vadim-PC:~/Repos/linux-6.18.33$ make mrproper
vadim@vadim-PC:~/Repos/linux-6.18.33$ ARCH=arm make defconfig
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/kconfig/conf.o
  HOSTCC  scripts/kconfig/confdata.o
  HOSTCC  scripts/kconfig/expr.o
  LEX     scripts/kconfig/lexer.lex.c
  YACC    scripts/kconfig/parser.tab.[ch]
  HOSTCC  scripts/kconfig/lexer.lex.o
  HOSTCC  scripts/kconfig/menu.o
  HOSTCC  scripts/kconfig/parser.tab.o
  HOSTCC  scripts/kconfig/preprocess.o
  HOSTCC  scripts/kconfig/symbol.o
  HOSTCC  scripts/kconfig/util.o
  HOSTLD  scripts/kconfig/conf
*** Default configuration is based on 'multi_v7_defconfig'
#
# configuration written to .config
#
vadim@vadim-PC:~/Repos/linux-6.18.33$
```

Собираем ядро:

```
vadim@vadim-PC:~/Repos/linux-6.18.33$ ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf
- make -j$(nproc) zImage
  SYNC    include/config/auto.conf.cmd
*
* Restart config...
*
*
* Kernel Features
*
Symmetric Multi-Processing (SMP) [Y/n/?] y
  Allow booting SMP kernel on uniprocessor systems (SMP_ON_UP) [Y/n/?] y
Support cpu topology definition (ARM_CPU_TOPOLOGY) [Y/n/?] y
Architected timer support (HAVE_ARM_ARCH_TIMER) [Y/?] y
Multi-Cluster Power Management (MCPM) [Y/?] y
big.LITTLE support (Experimental) (BIG_LITTLE) [N/y/?] n
Memory split
> 1. 3G/1G user/kernel split (VMSPLIT_3G)
  2. 3G/1G user/kernel split (for full 1G low memory) (VMSPLIT_3G_OPT)
  3. 2G/2G user/kernel split (VMSPLIT_2G)
  4. 1G/3G user/kernel split (VMSPLIT_1G)
choice[1-4?]: 1
Maximum number of CPUs (2-32) (NR_CPUS) [16] 16
Support for hot-pluggable CPUs (HOTPLUG_CPU) [Y/?] y
Support for the ARM Power State Coordination Interface (PSCI) (ARM_PSCI) [Y/?] y
Timer frequency
> 1. 100 Hz (HZ_100)
  2. 200 Hz (HZ_200)
  3. 250 Hz (HZ_250)
  4. 300 Hz (HZ_300)
  5. 500 Hz (HZ_500)
  6. 1000 Hz (HZ_1000)
choice[1-6?]: 1
Compile the kernel in Thumb-2 mode (THUMB2_KERNEL) [N/y/?] n
```

Сборка заняла примерно 4 минуты:

```
  Kernel: arch/arm/boot/Image is ready
  LDS     arch/arm/boot/compressed/vmlinux.lds
  AS      arch/arm/boot/compressed/head.o
  GZIP    arch/arm/boot/compressed/piggy_data
  CC      arch/arm/boot/compressed/misc.o
  CC      arch/arm/boot/compressed/decompress.o
  CC      arch/arm/boot/compressed/string.o
  AS      arch/arm/boot/compressed/hyp-stub.o
  CC      arch/arm/boot/compressed/fdt_rw.o
  CC      arch/arm/boot/compressed/fdt_ro.o
  CC      arch/arm/boot/compressed/fdt_wip.o
  CC      arch/arm/boot/compressed/fdt.o
  CC      arch/arm/boot/compressed/atags_to_fdt.o
  CC      arch/arm/boot/compressed/fdt_check_mem_start.o
  AS      arch/arm/boot/compressed/lib1funcs.o
  AS      arch/arm/boot/compressed/ashldi3.o
  AS      arch/arm/boot/compressed/bswapsdi2.o
  AS      arch/arm/boot/compressed/piggy.o
  LD      arch/arm/boot/compressed/vmlinux
  OBJCOPY arch/arm/boot/zImage
  Kernel: arch/arm/boot/zImage is ready
vadim@vadim-PC:~/Repos/linux-6.18.33$ cd arch/arm/boot
vadim@vadim-PC:~/Repos/linux-6.18.33/arch/arm/boot$ ls -ld zImage
-rwxrwxr-x 1 vadim vadim 11801088 May 29 15:29 zImage
vadim@vadim-PC:~/Repos/linux-6.18.33/arch/arm/boot$
```

Собираем device tree blob'ы:

```
vadim@vadim-PC:~/Repos/linux-6.18.33$ ARCH=arm make -j$(nproc) dtbs
  SYNC    include/config/auto.conf.cmd
  DTC     arch/arm/boot/dts/actions/owl-s500-cubieboard6.dtb
  DTC     arch/arm/boot/dts/airoha/en7523-evb.dtb
  DTC     arch/arm/boot/dts/amazon/alpine-db.dtb
  DTC     arch/arm/boot/dts/amlogic/meson8-minix-neo-x8.dtb
  DTC     arch/arm/boot/dts/arm/vexpress-v2p-ca5s.dtb
  DTC     arch/arm/boot/dts/axis/artpec6-devboard.dtb
  DTC     arch/arm/boot/dts/calxeda/highbank.dtb
  DTC     arch/arm/boot/dts/aspeed/aspeed-ast2500-evb.dtb
  DTC     arch/arm/boot/dts/broadcom/bcm2835-rpi-b.dtb
  DTC     arch/arm/boot/dts/broadcom/bcm2835-rpi-a.dtb
  DTC     arch/arm/boot/dts/allwinner/sun4i-a10-a1000.dtb
  DTC     arch/arm/boot/dts/cnxt/cx92755_equinox.dtb
  DTC     arch/arm/boot/dts/actions/owl-s500-guitar-bb-rev-b.dtb
  DTC     arch/arm/boot/dts/allwinner/sun4i-a10-ba10-tvbox.dtb
  DTC     arch/arm/boot/dts/aspeed/aspeed-ast2600-evb-a1.dtb
  DTC     arch/arm/boot/dts/arm/vexpress-v2p-ca9.dtb
  DTC     arch/arm/boot/dts/calxeda/ecx-2000.dtb
  DTC     arch/arm/boot/dts/amlogic/meson8-fernsehfee3.dtb
  DTC     arch/arm/boot/dts/aspeed/aspeed-ast2600-evb.dtb
  DTC     arch/arm/boot/dts/actions/owl-s500-labrador-base-m.dtb
```

Переносим ядро и dtb-файл для vexpress-a9 в отдельную директорию:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ ls -l
total 11544
-rw-rw-r-- 1 vadim vadim    14329 May 29 15:39 vexpress-v2p-ca9.dtb
-rwxrwxr-x 1 vadim vadim 11801088 May 29 15:29 zImage
vadim@vadim-PC:~/Emulation/arm-vexpress$
```

Запускаем полученную прошивку (без корневой фаловой системы) через эмулятор QEMU:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic
[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 6.18.33 (vadim@vadim-PC) (arm-linux-gnueabihf-gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1 SMP Fri May 29 15:29:39 +07 2026
[    0.000000] CPU: ARMv7 Processor [410fc090] revision 0 (ARMv7), cr=10c5387d
[    0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT nonaliasing instruction cache
[    0.000000] OF: fdt: Machine model: V2P-CA9
[    0.000000] Memory policy: Data cache writeback
[    0.000000] efi: UEFI not found.
[    0.000000] cma: Failed to reserve 64 MiB
[    0.000000] Zone ranges:
[    0.000000]   DMA      [mem 0x0000000060000000-0x0000000067ffffff]
[    0.000000]   Normal   empty
[    0.000000]   HighMem  empty
[    0.000000] Movable zone start for each node
[    0.000000] Early memory node ranges
[    0.000000]   node   0: [mem 0x0000000060000000-0x0000000067ffffff]
[    0.000000] Initmem setup node 0 [mem 0x0000000060000000-0x0000000067ffffff]
```

Ядро успешно запускается, но спустя 2 секунды получаем падение ядра:

```
[    2.060456]  squashfs
[    2.060459]  vfat
[    2.060462]  msdos
[    2.060465]  ntfs
[    2.060468]  ntfs3
[    2.060475]
[    2.060651] Kernel panic - not syncing: VFS: Unable to mount root fs on unkno
wn-block(0,0)
[    2.064480] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Not tainted 6.18.33 #1 NONE
[    2.064746] Hardware name: ARM-Versatile Express
[    2.064956] Call trace:
[    2.065439]  unwind_backtrace from show_stack+0x10/0x14
[    2.066226]  show_stack from dump_stack_lvl+0x54/0x68
[    2.066414]  dump_stack_lvl from vpanic+0xc4/0x2f0
[    2.066563]  vpanic from __do_trace_suspend_resume+0x0/0x4c
[    2.067292] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs
 on unknown-block(0,0) ]---
```